

# SAAMYUKKTH SURESH

## Software Development Engineer | Backend Systems, Python & FastAPI

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### PROFESSIONAL SUMMARY

Software engineer with 1.5 years at Prodapt (including internship) building production backend services in Python and FastAPI: async REST and WebSocket APIs, microservices, relational and vector data pipelines, and containerized deployments on Azure. Has shipped several enterprise platforms end to end, from data modeling and API design to real-time streaming, CI/CD, and integration of AI services.

### TECHNICAL SKILLS

**Languages:** Python, C++, SQL, TypeScript, JavaScript

**Backend:** FastAPI, REST APIs, WebSockets, SSE, async programming, microservices, Pydantic, authentication/authorization

**Data:** MySQL, Redis, Milvus (vector DB), Neo4j (graph DB), data modeling

**Cloud & DevOps:** Microsoft Azure, Docker, CI/CD, Git/GitHub, Postman, unit testing, Agile/Scrum, Jira

**Frontend:** React, Next.js, TypeScript

**AI Integration:** Azure OpenAI, Azure AI Speech, LangGraph, LangChain, RAG, embeddings and vector search, Model Context Protocol (MCP)

**CS Fundamentals:** Data structures and algorithms, system design, object-oriented design

### PROFESSIONAL EXPERIENCE

#### Software Development Engineer, Prodapt, Jul 2025 – Present

Chennai, Tamil Nadu

- Built the backend for an end-to-end **AI Interview Platform** as FastAPI microservices covering job-description generation, candidate-to-JD matching, interview planning, and live interview sessions, with MySQL for core records and Milvus for embedding-based matching on skills and work experience.
- Engineered a **real-time interview streaming pipeline** over WebSockets that integrates Azure AI Speech (speech-to-text and neural TTS with SSML voice control) with LLM answer analysis (GPT-5 mini), maintaining per-session interview state so follow-up questions adapt to prior answers.
- Architected backend pipelines for **SKILLOS**, a skill-intelligence platform (Azure OpenAI embeddings + Milvus) powering semantic skill extraction, vector similarity search, and competency-gap analytics across 1,000+ employee profiles.
- Engineered an assessment engine on FastAPI microservices and MySQL with dynamic question generation and automated scoring, reducing manual assessment effort by ~60%.
- Built and deployed employee-attrition prediction models (Scikit-learn, SHAP; ROC-AUC 0.65) and exposed interpretable, feature-level insights to HR stakeholders through REST APIs.

#### Software Development Engineer Intern, Prodapt, Jan 2025 – Jul 2025

Chennai, Tamil Nadu

- Designed and shipped a production conversational AI service (LangChain, LangGraph, Azure OpenAI, Milvus) with async REST endpoints, document retrieval, and persistent multi-turn conversation memory.
- Built asynchronous multi-agent workflows (CrewAI) for news aggregation and summarization, cutting manual content-processing time by ~70%.

### PROJECTS

#### Lumora: AI-Powered Personal Knowledge Management System - Python, FastAPI, React, TypeScript, Neo4j, Docker

[github.com/saamyukkth26/Lumora](https://github.com/saamyukkth26/Lumora) | Live: [lumora-rust-three.vercel.app](https://lumora-rust-three.vercel.app)

- Built a full-stack knowledge-management app: FastAPI backend with real-time SSE streaming chat, React + TypeScript frontend, and a hybrid search pipeline (BM25 + dense embeddings with re-ranking) over PDFs, notes, and web content.
- Integrated 4 Model Context Protocol (MCP) tool servers (filesystem, web search, arXiv, Wikipedia) behind LangGraph agents with a critique-and-retry loop for multi-step research tasks.
- Modeled extracted entities as a Neo4j knowledge graph with an interactive React Flow visualization; containerized the full stack with Docker and deployed on Vercel and Render.

### EDUCATION

#### B.Tech, Computer Science (Specialization in Artificial Intelligence & Machine Learning), 2021 – 2025

Vellore Institute of Technology (VIT), Chennai | CGPA: 8.65 / 10

### CERTIFICATIONS

- Claude Certified Developer – Foundations**, Anthropic: Claude API, Claude Code, agent development, MCP, tool use.
- Claude Certified Architect – Foundations**, Anthropic: agentic architecture, multi-agent orchestration, tool integration.
- LLM Engineering: Master AI, LLMs and Agents (Udemy) | LangChain & LangGraph LLM Applications (Udemy)